DATA CENTER SIMULATION PERFORMANCE REPORT

Profile: NORMAL_DAY | Generated: 2026-08-23 18:35:55

Parameter	Value
Experiment Duration	600.0 seconds
Start Time (T1)	1787489753.9102137
End Time (T2)	1787490353.9102137
Primary Bottleneck	None
Confidence Level	HIGH
Failure Classification	HTTP_REQUEST_FAILURES

SLA Metric	Measured Value	Target Limit	Result
P95 Response Time	110.0 ms	500.0 ms	PASS
Error Rate	1.0 %	1.0 %	PASS
Peak CPU Utilization	2.86 %	80.0 %	PASS
Peak Memory Utilization	17.55 %	90.0 %	PASS

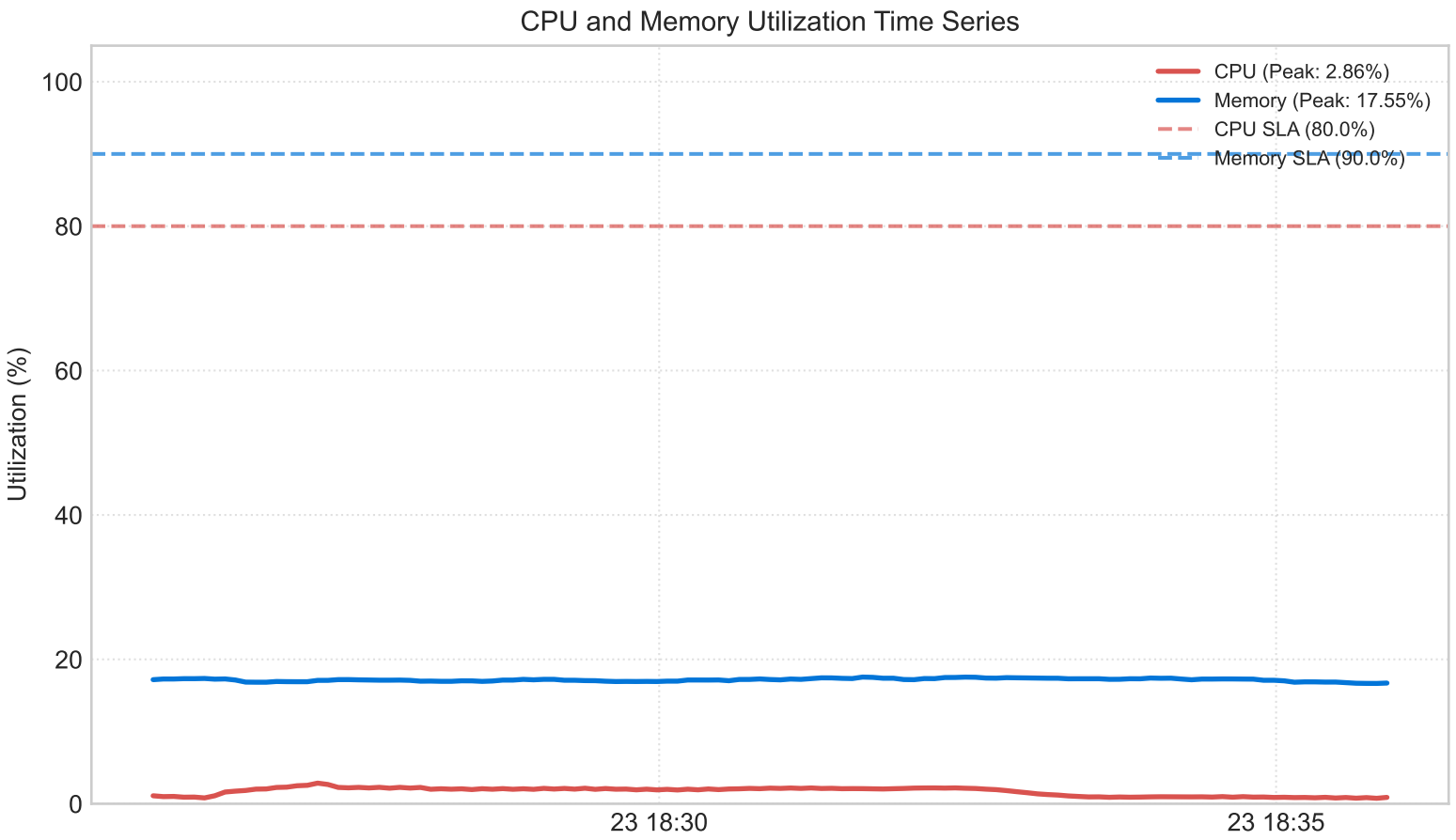
Executive Conclusion & Evidence Synthesis:

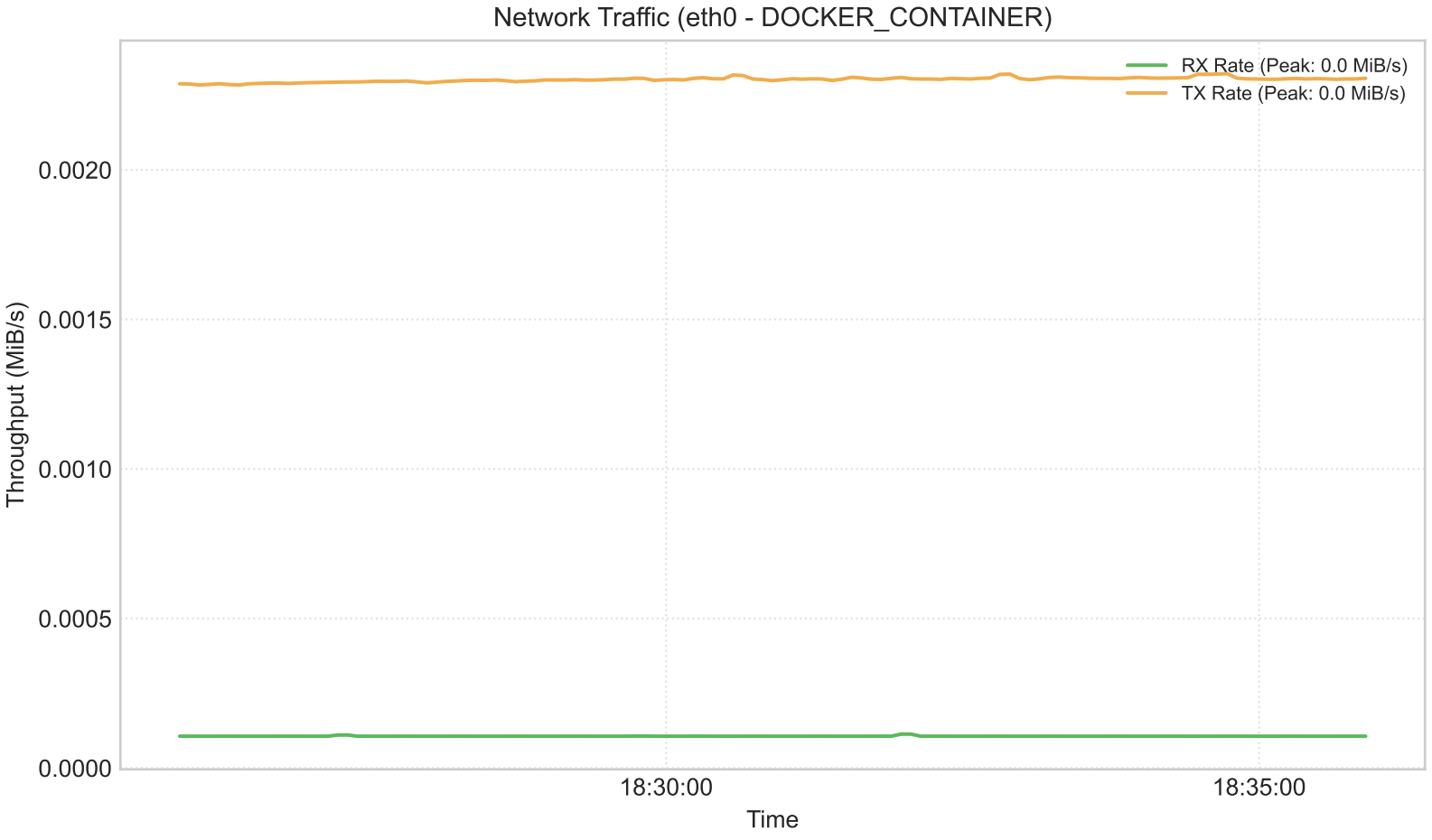
The experiment executed 8275 requests with overall SLA status 'PASS' (4/4 checks satisfied). No container crashes or service interruptions were detected by the watchdog. The system operated within all performance, reliability, and resource boundaries.

Diagnostic Observations:

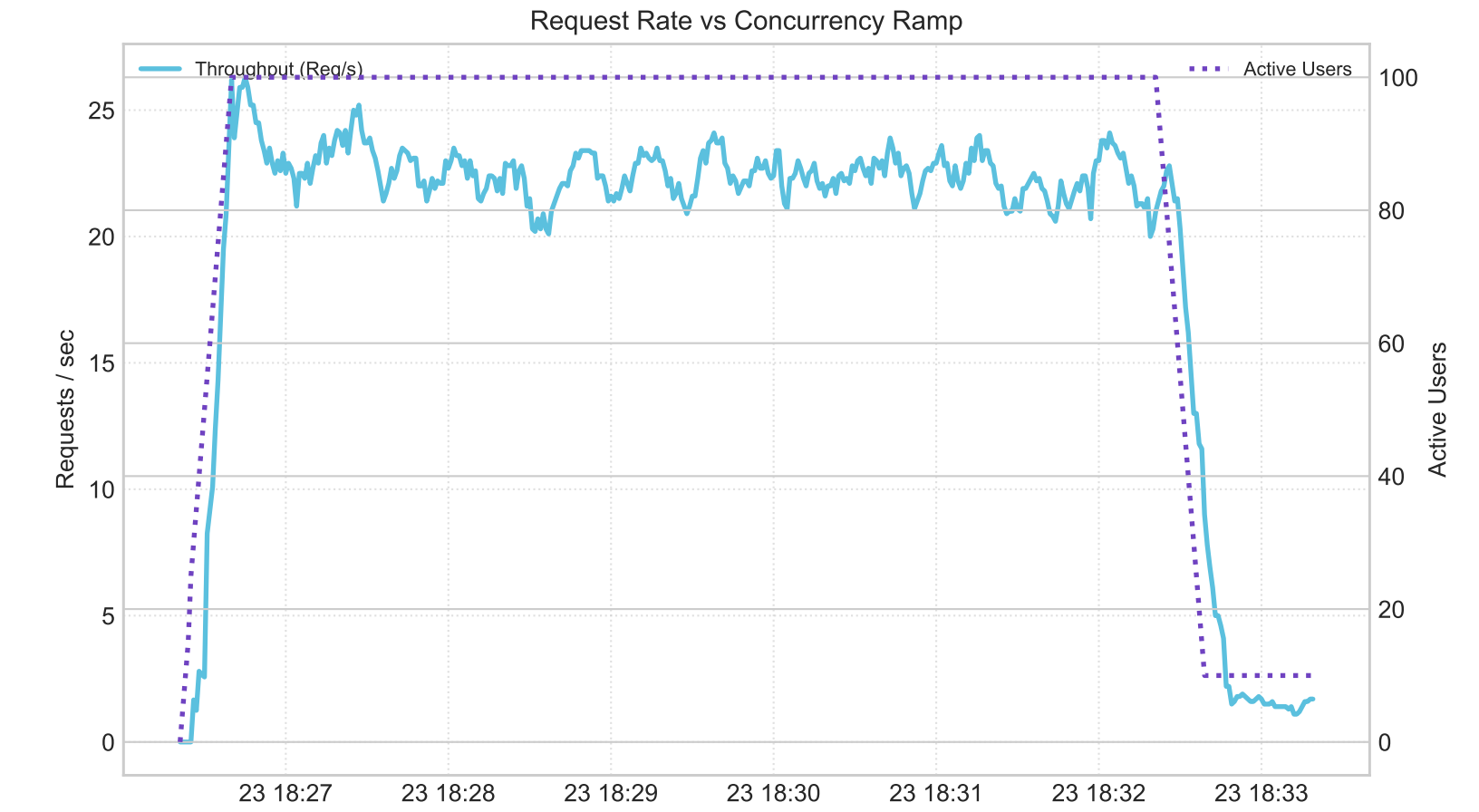
• All measured system and load parameters remained within safe thresholds.

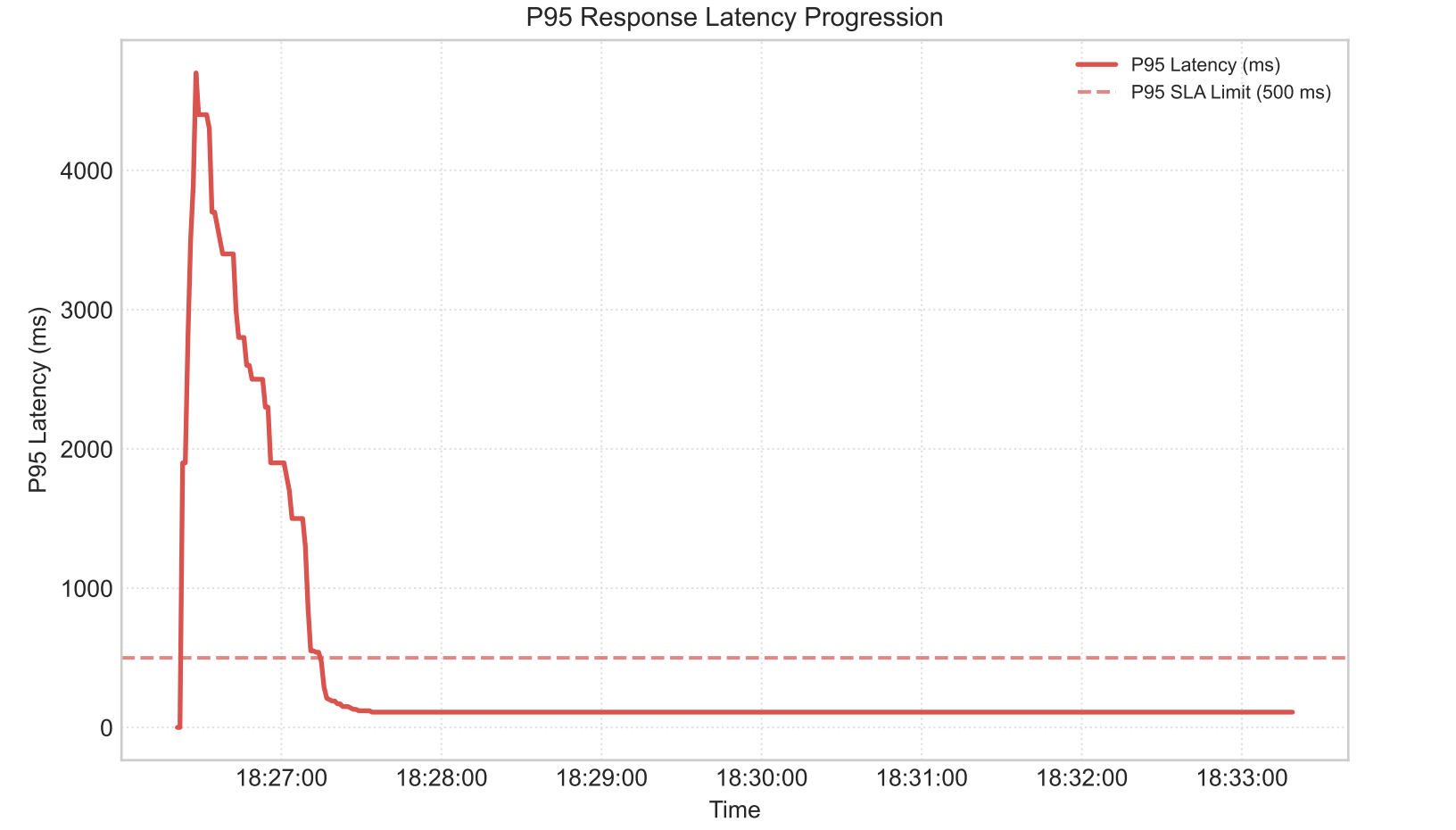
System & Hardware Resource Dynamics





Locust Workload & Latency Profile





Endpoint Breakdown & Chronological Timeline

HTTP Endpoint Performance Metrics:

Endpoint	Requests	Fails	Avg Latency	P95 Latency	RPS
GET /api/data	2621	0	75.61 ms	120.0 ms	6.24
GET /health	3483	2	13.64 ms	31.0 ms	8.29
POST /login	100	80	1106.68 ms	4400.0 ms	0.24
POST /logout	19	0	29.79 ms	190.0 ms	0.05
GET /me	1664	1	21.51 ms	33.0 ms	3.96
POST /profile/update	388	0	91.15 ms	110.0 ms	0.92

Chronological Event Timeline:

Time	Event Type	Category	Severity	Description
[RUNNER] 18:25:53	EXPERIMENT_START	OBSERVED	INFO	Experiment started at 1787489753.9102137)
18:26:21	[LOCUST] LOAD_STARTED	OBSERVED	INFO	Workload generation began with 0 users
[LOCUST] 18:26:40	LOCUST_PEAK_USERS	OBSERVED	INFO	Locust concurrency reached peak of 100 active...
[LOCUST] 18:26:45	LOCUST_PEAK_RPS	OBSERVED	INFO	Locust throughput reached peak of 26.3 req/s
[PROMETHEUS] 18:27:13	PROMETHEUS_CPU_PEAK	OBSERVED	INFO	CPU utilization peaked at 2.86%
[PROMETHEUS] PROMETHEUS_MEMORY_PEAK	OBSERVED	INFO	Memory utilization peaked at 17.55%	
[PROMETHEUS] 18:27:58	PROMETHEUS_NETWORK_PEAK_RX	OBSERVED	INFO	Network receive rate peaked at 0.0 MiB/s on e...
[PROMETHEUS] 18:27:43	PROMETHEUS_NETWORK_PEAK_TX	OBSERVED	INFO	Network transmit rate peaked at 0.0 MiB/s on ...
[RUNNER] 18:39:53	EXPERIMENT_END	OBSERVED	INFO	Experiment completed at 1787490353.910213...